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6.2. a. $\Pi_{person_name}(\sigma_{city="Miami"}(employee))$

b. $\Pi_{person_name}(\sigma_{salary>100000}(works))$

c. $\Pi_{person_name}(\sigma_{city="Miami} \wedge_{salary>100000}(employee \bowtie_{employee.person_name = works.person_name} works)$

6.3

a. $\Pi_{branch_name}(\sigma_{branch_city="Chicago}(branch))$

b. $\Pi_{ID}(\sigma_{branch_name="Dawntown"}(borrower \bowtie_{borrower.loan_number = loan.loan_number} loan))$

6.4.

a. $\Pi_{ID, person_name}(employee) - \Pi_{ID, person_name}(\sigma_{works.company_name="BigBank"}(employee \bowtie_{employee.person_name = works.person_name} works))$

b. $\Pi_{ID, person_name}(\sigma_{salary > \Pi_{salary}(works)}(employee \bowtie_{employee.person_name} works))$

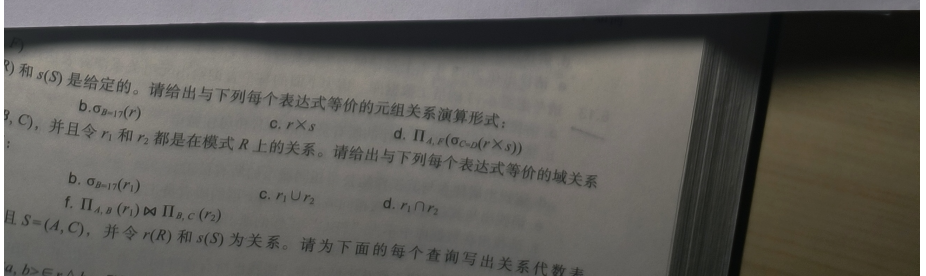
6.6 a. $\{t \mid \exists u \in r(u[A] = t[A])\}$

b. $\{t \mid t \in r \wedge t[B] = 1\}$

c. $\{t \mid \exists u \in r(u[A] = t[A] \wedge u[B] = t[B] \wedge u[C] = t[C]) \wedge \exists v \in s(v[D] = t[D] \wedge v[E] = t[E] \wedge v[F] = t[F])\}$

d. $\{t \mid \exists u \in r(u[A] = t[A]) \wedge \exists v \in s(v[F] = t[F]) \wedge u[C] = v[D]\}$

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$r(R)$ 和 $s(S)$ 是给定的。请给出与下列每个表达式等价的元组关系演算形式：

b. $\sigma_{B=17}(r)$　　c. $r \times s$　　d. $\Pi_{A,B}(\sigma_{C=D}(r \times s))$

(A, C)，并且令 r_1 和 r_2 都是在模式 R 上的关系。请给出与下列每个表达式等价的域关系

b. $\sigma_{B=17}(r_1)$　　c. $r_1 \cup r_2$　　d. $r_1 \cap r_2$

f. $\Pi_{A,B}(r_1) \bowtie \Pi_{B,C}(r_2)$

且 $S=(A, C)$，并令 $r(R)$ 和 $s(S)$ 为关系。请为下面的每个查询写出关系代数表

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6.8 . a. $\Pi_a(\sigma_{b=7}(r))$

b. $r \bowtie_{r.a=s.a} s$

c. $\Pi_a(\sigma_{m.b_1>m.b_2}(\rho_m(\rho_{m_1(a,b_1)}(\Pi_{a,b}(\sigma_{r.a=s.a}(r\times s))) \bowtie_{m_1.a=m_2.a} \rho_{m_2(a,b_2)}(\Pi_{a,b}(\sigma_{r.a=s.c}(r\times s))))))$

6.10 .

a. $\Pi_{ID, person_name}(\sigma_{company_name="BigBank"}(employee \bowtie_{employee.person_name = works.person_name} works))$

b. $\Pi_{ID, person_name, city}(\sigma_{company_name="BigBank"}(employee \bowtie_{employee.person_name = works.person_name} works))$

c. $\Pi_{ID, person_name, street, city}(\sigma_{company_name="BigBank" \wedge salary>10000}(employee \bowtie_{employee.person_name = works.person_name} works))$

d. $\Pi_{ID, person_name}((employee \bowtie_{employee.person_name = work.person_name} works) \bowtie_{employee.city = company.city} company)$

6.11

a. $\Pi_{loan_numer}(\sigma_{amount>10000}(loan))$

b. $\Pi_{ID}(\sigma_{balance>6000}(depositor \bowtie_{depositor.account_number = account.account_number} account))$

c. $\Pi_{ID}(\sigma_{branch_name="Uptown" \wedge balance>6000}(depositor \bowtie_{depositor.account_number = account.account_number} account))$

, B, C)
, E, F)
系 $r(R)$ 和 $s(S)$ 是给定的。请给出与下列每个表达式等价的元组关系演算形式：
(r) b. $\sigma_{B=17}(r)$ c. $r \times s$ d. $\Pi_{A,F}(\sigma_{C=D}(r \times s))$
$= (A, B, C)$，并且令 r_1 和 r_2 都是在模式 R 上的关系。请给出与下列每个表达式等价的域关系
表达式：
(r_1) b. $\sigma_{B=17}(r_1)$ c. $r_1 \cup r_2$ d. $r_1 \cap r_2$
$- r_2$ f. $\Pi_{A,B}(r_1) \bowtie \Pi_{B,C}(r_2)$
$= (A, B)$ 且 $S = (A, C)$，并令 $r(R)$ 和 $s(S)$ 为关系。请为下面的每个查询写出关系代数表

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6.13

a. $\{t \mid \exists s \in employee(t[person_name] = s[person_name] \wedge \exists u \in works(s[person_name] = u[person_name] \wedge u[company_name] = "FB("))\}$

b. $\{t \mid \exists s \in employee(t[person_name] = s[person_name] \wedge t[city] = s[city] \wedge \exists u \in works(s[person_name] = u[person_name] \wedge u[company_name] = "FB("))\}$

c. $\{t \mid \exists s \in employee(t[person_name] = s[person_name] \wedge t[street] = s[street] \wedge t[city] = s[city] \wedge \exists u \in works(s[person_name] = u[person_name] \wedge u[salary] > 10000 \wedge u[company_name] = "FB("))\}$

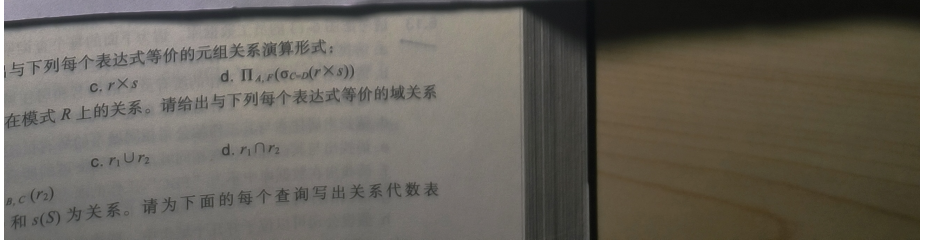
d.

$\{t \mid t \in employee(\exists u \in works(t[person_name] = u[person_name] \wedge \exists v \in company(t[city] = v[city])))\}$

e.

$\{t \mid t \in employee(\exists u \in employee(\exists v \in manages(u[person_name] = v[person_name] \wedge t[city] = u[city] \wedge t[street] = u[street])))\}$

~~f. $\{t \mid t \in employee \wedge \neg \exists u \in employee(t[person_name] = u[person_name] \wedge u[city] =$~~



f. $\{t \mid t \in employee \wedge (\exists u \in works(t[person_name] = u[person_name] \wedge \neg\exists v \in employee\, t \neg \exists v \in works(t[person_name] = v[person_name] \wedge v[company_name] = "FB")))\}$

g. $\{t \mid t \in employee \wedge (\neg\exists u \in works(t[person_name] = u[person_name] \wedge \neg\exists v \in works(v[company_name] = "SB" \wedge v[salary] \geq u[salary])))\}$

h. $\{t \mid t \in company \wedge (\forall u \in company(u[name] = "SB" \wedge \exists m \in company(m[company_name] = t[company_name] \wedge m[city] = u[city])))\}$